

Investigation

Time limit: 1.00 s

Memory limit: 512 MB

You are going to travel from Syrjälä to Lehmälä by plane. You would like to find answers to the following questions:

- what is the minimum price of such a route?
- how many minimum-price routes are there? (modulo $10^9 + 7$)
- what is the minimum number of flights in a minimum-price route?
- what is the maximum number of flights in a minimum-price route?

Input

The first input line contains two integers n and m : the number of cities and the number of flights. The cities are numbered $1, 2, \dots, n$. City 1 is Syrjälä, and city n is Lehmälä.

After this, there are m lines describing the flights. Each line has three integers a, b , and c : there is a flight from city a to city b with price c . All flights are one-way flights.

You may assume that there is a route from Syrjälä to Lehmälä.

Output

Print four integers according to the problem statement.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Example

Input	Output
4 5 1 4 5 1 2 4 2 4 5 1 3 2 3 4 3	5 2 1 2